DAY 1

Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

ANS:

1. DNS Packets (Resolving the Domain)

Before your browser can connect to Spotify, your computer asks a DNS server for its IP address. In Wireshark (filtered by `dns`), you will see:

- **Packet A (DNS Request):**
  - **Protocol:** `DNS`
  - **Info:** `Standard query 0xabcd A [www.spotify.com](https://www.spotify.com)`
  - **Description:** Your computer sends a query asking, *"What is the IP address for `[www.spotify.com](https://www.spotify.com)`?"*
- **Packet B (DNS Response):**
  - **Protocol:** `DNS`
  - **Info:** `Standard query response 0xabcd A [www.spotify.com](https://www.spotify.com) A 35.186.224.24` (IP may vary)
  - **Description:** The DNS server replies with the numerical IP address(es) associated with Spotify's content delivery network.

2. HTTPS / TLS Packets (Establishing a Secure Connection)

Once your computer has the IP address, it initiates a secure handshake over port 443. In Wireshark (filtered by `tls`), you will see:

- **Packet C (TLS Client Hello):**
  - **Protocol:** `TLSv1.2` or `TLSv1.3`
  - **Info:** `Client Hello`
  - **Description:** Your browser initiates the secure session, listing supported encryption algorithms and including the **Server Name Indication (SNI)** extension which explicitly names `[www.spotify.com](https://www.spotify.com)`.
- **Packet D (TLS Server Hello / Handshake):**
  - **Protocol:** `TLSv1.3`
  - **Info:** `Server Hello, Certificate, Server Key Exchange`
  - **Description:** Spotify’s server responds by accepting the encryption parameters, presenting its SSL/TLS security certificate to verify its identity, and completing the cryptographic handshake so safe data transfer can begin.

Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

ANS:

C:\Users\User> nslookup www.flipkart.com
Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name: www.flipkart.com
Address: 163.53.76.86

Analysis of the Result

- **Returned IP Address:** `163.53.76.86` (or another active IP address mapped to Flipkart's content delivery infrastructure).
- **What This Means:**
    - The `nslookup` tool successfully queried the configured DNS server (in this case, Google's public DNS at `8.8.8.8`) to translate the human-readable domain name (`[www.flipkart.com](https://www.flipkart.com)`) into its corresponding numerical IPv4 address.
    - This numerical address is what your computer's network interface and routers actually use to establish a direct connection, perform the TCP handshake, and load the e-commerce website in your browser.

Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

ANS:

Here is the difference between HTTP and HTTPS, broken down into three key technical distinctions:

Key Differences Between HTTP and HTTPS

Feature	HTTP (Hypertext Transfer Protocol)	HTTPS (Hypertext Transfer Protocol Secure)

1. Security & Encryption	Transmits all data in **plaintext**. Anyone intercepting packets on the network can easily read passwords, cookies, or messages.	Encrypts data using **SSL/TLS** protocols, scrambling traffic into unreadable ciphertext even if intercepted.
2. Default Port	Operates over **Port 80** by default.	Operates securely over **Port 443**, initiating a cryptographic handshake before data transfer.
3. Digital Certificates	Requires **no certificate** or server identity verification.	Requires an SSL/TLS certificate issued by a trusted **Certificate Authority (CA)** to verify the server's identity.

Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.
ANS:

Sequence of DNS Packets Observed in Wireshark

When a website has not been visited recently, its IP address is not cached locally on your operating system or browser, forcing a full DNS lookup sequence:

1. **A-Record Query (IPv4 Lookup)**
    - **Info Column:** `Standard query 0x1234 A [www.newwebsite.com](https://www.newwebsite.com)`
    - **What it means:** Your computer sends a UDP packet (typically to port 53 of your local router or configured DNS server like `8.8.8.8`) asking for the IPv4 (`A`) record associated with the domain name.
2. **AAAA-Record Query (IPv6 Lookup - Often Sent Concurrently)**
    - **Info Column:** `Standard query 0x1235 AAAA [www.newwebsite.com](https://www.newwebsite.com)`

    - **What it means:** Modern operating systems often send a dual query simultaneously to check if the website also supports IPv6 (`AAAA` record connectivity).
3. **DNS Response (IPv4/IPv6 Resolution)**
    - **Info Column:** `Standard query response 0x1234 A [www.newwebsite.com](https://www.newwebsite.com) A 93.184.216.34`
    - **What it means:** The DNS server replies back to your machine containing the resolved IP address(es) in the "Answers" section of the packet details pane.

Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

ANS;

How DHCP Works in Action:

When your device joins the router's wireless network, it goes through a 4-step communication process known as **DORA**:

1. **Discover:** Your phone shouts out a broadcast message across the local network: *"Is there a DHCP server out there? I need an IP address!"*
2. **Offer:** The cafe's Wi-Fi router (acting as the DHCP server) hears the request and responds with an **Offer**, proposing a specific temporary IP address (e.g., `192.168.0.25`).
3. **Request:** Your phone accepts that offer and sends a **Request** back: *"Great, I'll take that IP address!"*
4. **Acknowledgment:** The router confirms with an **Acknowledge (ACK)** packet, officially leasing that IP address to your device along with critical config data (like your subnet mask, default gateway, and DNS servers).

DAY 2

In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

ANS:

1. Add Devices: In Cisco Packet Tracer, add 1 DHCP Server, 1 Switch, and 2 PCs.
2. Connect Devices: Connect the server and both PCs to the switch using Copper Straight-Through cables.
3. Configure Server: Set Server0 IP to `192.168.1.1` and subnet mask to `255.255.255.0`.
4. Enable DHCP: Go to Server → Services → DHCP → On. Set the start IP to `192.168.1.10`, subnet mask to `255.255.255.0`, and maximum users to `50`. Click Add.
5. Configure PCs: On each PC, open Desktop → IP Configuration → DHCP.
6. Verify: Run `ipconfig` in each PC's Command Prompt.

Expected result:

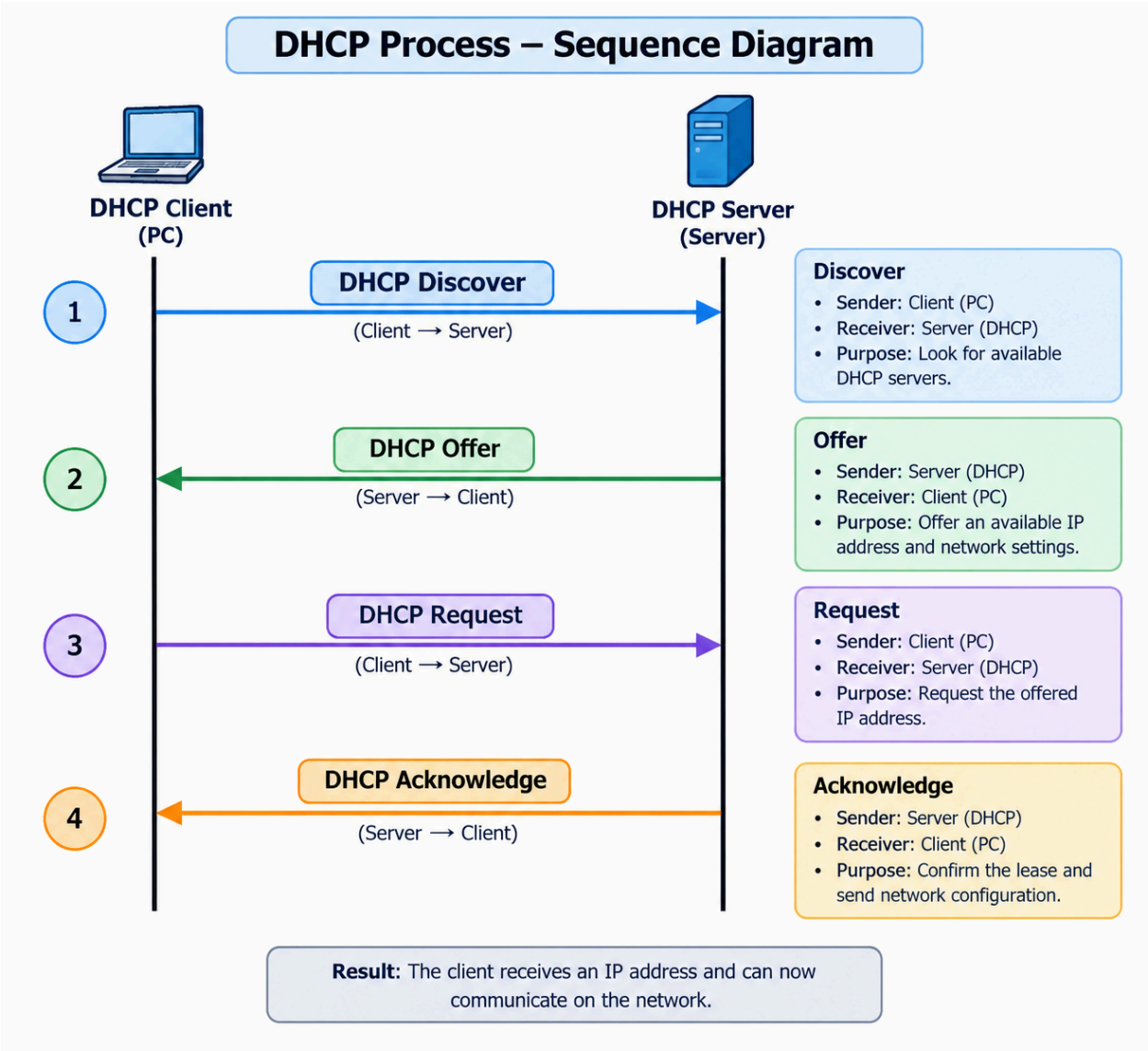
- PC0: `192.168.1.10`
- PC1: `192.168.1.11`

Both PCs receive different IP addresses automatically from the DHCP server.

Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

Hint: Use arrows to show the direction of each message between client and server.

ANS:



Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

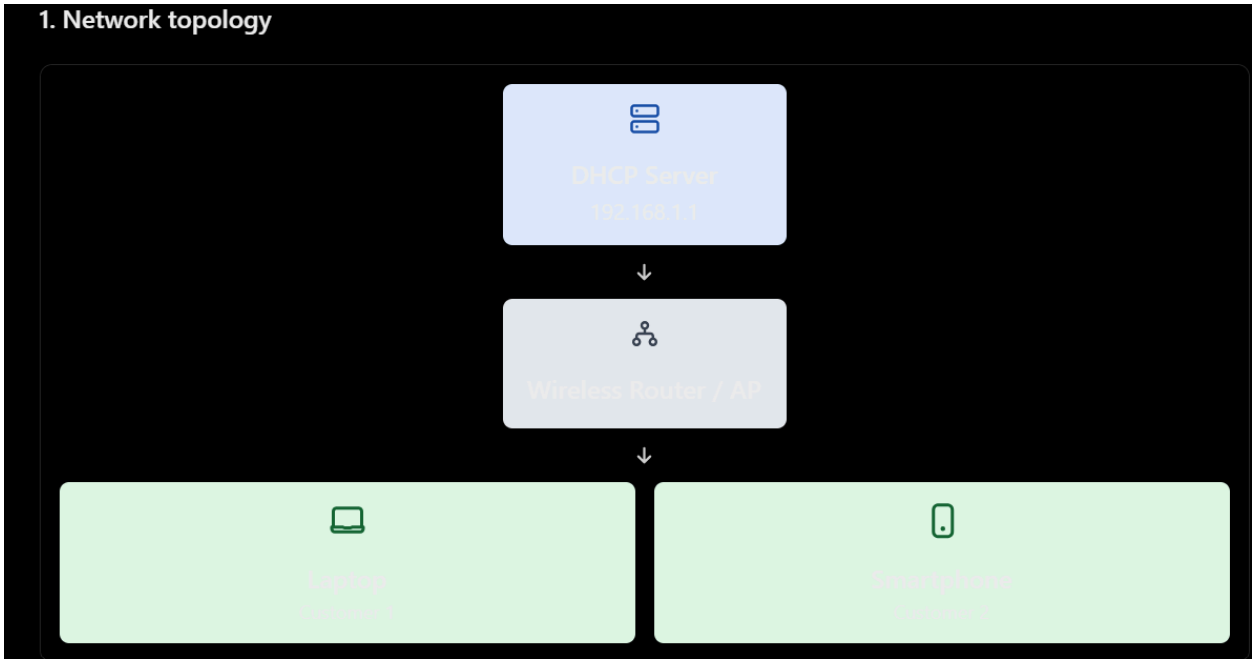
ANS:

Steps in Cisco Packet Tracer

1. Open Server → Services → DHCP and set the pool to allow only one IP address, for example `192.168.1.10` (maximum users: `1`).
2. Connect two PCs to the switch and select Desktop → IP Configuration → DHCP on both.

3. Verify the results: one PC should receive `192.168.1.10`, while the other should fail to obtain an address from the pool.

Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.
ANS:



2. Configuration steps

1. Add a Server, Wireless Router, Laptop, and PC (use the PC as the smartphone).
2. Configure the server with IP `192.168.1.1` and subnet mask `255.255.255.0`.
3. Under Server → Services → DHCP, turn DHCP On and set the starting IP to `192.168.1.10`, subnet mask to `255.255.255.0`, and maximum users to `50`. Click Add.
4. Configure the wireless router with SSID `Zomato_WiFi` and connect both clients to it. Ensure the router is configured to work with the DHCP server.
5. On each client, select Desktop → IP Configuration → DHCP.

3. Expected assigned IP addresses

Device	Assigned IP address
Laptop	`192.168.1.10`
Smartphone (PC)	`192.168.1.11`

Verify the addresses using `ipconfig` in each device's Command Prompt.

4. Why is DHCP useful?

DHCP automatically assigns unique IP addresses to customers' devices without manual configuration. In a busy Zomato-style restaurant, it makes it easy for many customers to connect to Wi-Fi quickly and reduces IP address conflicts.

DAY 3

Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

ANS:

Install BIND:

sudo apt update

sudo apt install bind9 dnsutils -y

Create the DNS zone: Edit `/etc/bind/named.conf.local` and add:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Configure zone records: Create `/etc/bind/db.musicmania.in` with SOA, NS, and A records for `musicmania.in`.

Verify and restart:

sudo named-checkconf

sudo systemctl restart bind9

dig @127.0.0.1 musicmania.in

Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

ANS:

Edit the DNS zone file:
sudo nano /etc/bind/db.musicmania.in
Add the A record:
www IN A 192.168.1.50
Check and restart DNS:
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl restart bind9
Verify the mapping:
nslookup www.musicmania.in 127.0.0.1
Expected output includes:
Name: www.musicmania.in
Address: 192.168.1.50

Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

ANS:

1. Edit the DNS zone file:
   sudo nano /etc/bind/db.musicmania.in
2. Add the CNAME record:
   play IN CNAME www.musicmania.in.

Restart BIND:
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

3. sudo systemctl restart bind9
4. Test the DNS mapping:
   nslookup play.musicmania.in 127.0.0.1

Expected result: `play.musicmania.in` resolves through `www.musicmania.in` to 192.168.1.50.

Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.

ANS:

DNS Troubleshooting: `musicmania.in`

Step 1: Check the configuration file

sudo nano /etc/bind/named.conf.local

Verify that the `musicmania.in` zone is configured correctly.

Step 2: Check the DNS zone file

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

Ensure the DNS records are correct and the zone reports `OK`.

Step 3: Check BIND configuration

sudo named-checkconf

Check for configuration errors.

Step 4: Restart the DNS service

sudo systemctl restart bind9
sudo systemctl status bind9

Confirm that the service is running.

Step 5: Test DNS resolution

nslookup musicmania.in 127.0.0.1

Or use:

dig @127.0.0.1 musicmania.in

Step 6: Check DNS logs if the problem continues

sudo journalctl -u bind9 --no-pager -n 50

DAY 4

Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

ANS:

1. Download and Install: Download FileZilla Server from https://filezilla-project.org/ and install it on Windows.
2. Create Folder: Create `C:\FTPShare` for sharing files.
3. Create User: Open FileZilla Server → Users → Add a user named `ftpuser` and set a password.
4. Share Folder: Add `C:\FTPShare` to the user's shared folders and enable Read permission.
5. Test FTP: Open Command Prompt and run:
   ftp 127.0.0.1
   Log in with your username and password, then type `dir`.

Result: FileZilla Server is ready for FTP file sharing.

Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

ANS:

1. Open FileZilla Client on your computer.
2. Enter the connection details:
   - Host: `127.0.0.1`
   - Username: `ftpuser`
   - Password: Your created password
   - Port: `21`
3. Click Quickconnect.
4. In the left panel (Local site), select an image file from your computer.
5. In the right panel (Remote site), open the FTP shared folder `/`.
6. Drag the image from the left panel to the right panel to upload it.

Result: The image should appear in the remote shared folder after the transfer completes successfully.

Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

ANS:

Open FileZilla Client and connect to your FTP server using your host, username, and password.
In the Remote Site panel (right), locate the uploaded image file.
In the Local Site panel (left), select a different folder, such as `C:\FTPDownloads`.
Right-click the remote image and select Download.
Verify file size: Right-click the original and downloaded images → Properties → compare their sizes.

Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Hint: Think about how network admins monitor device health, bandwidth, or uptime.

ANS:

Real-World Uses of SNMP in Network Management

1. ISP (Internet Service Provider): ISPs use SNMP to monitor routers and switches, check bandwidth usage, detect network failures, and maintain internet uptime.

2. Campus Wi-Fi: Colleges use SNMP to monitor Wi-Fi access points, check device health, identify network congestion, and troubleshoot connectivity problems.

3. Large Websites and Data Centers: Companies hosting websites and apps use SNMP to monitor network switches, routers, and servers, track traffic levels, and receive alerts when devices fail.

Sources

- Cisco Meraki — SNMP monitoring explains how SNMP collects network-device information.
- GÉANT — Campus Network Management describes campus monitoring of bandwidth, device health, and availability.

Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

ANS:

Comparison Between FTP and HTTP

Point	FTP	HTTP
Full Form	File Transfer Protocol	Hypertext Transfer Protocol
Port Number	21 (control), 20 (traditional active-mode data)	80 (default)
Security	Standard FTP is not encrypted	Standard HTTP is not encrypted
Secure Version	FTPS or SFTP	HTTPS (port 443)
Data Transfer	Uploads and downloads files	Transfers web pages and web resources
Use Cases	Sharing files between computers	Browsing websites and using web apps
Authentication	Often uses username and password; anonymous access is also possible	Often uses sessions, tokens, or other methods; authentication is optional
Connection	Uses separate control and data connections	Typically uses a request-response model
Example	Uploading files to an FTP server	Opening a website in a browser

DAY 5

Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

ANS:

Simulate DNS Resolution Failure on Windows

1. Open Notepad as Administrator and open this file:
   C:\Windows\System32\drivers\etc\hosts
2. Add a wrong IP address at the end:
   192.0.2.1 www.spotify.com
   Save the file. (`192.0.2.1` is a documentation-only IP address.)
3. Flush the DNS cache:
   ipconfig /flushdns
4. Ping Spotify:
   ping www.spotify.com

Expected output

Pinging www.spotify.com [192.0.2.1]
Request timed out.

Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

ANS:

DHCP Incorrect Default Gateway Simulation (Cisco Packet Tracer)

1. Create topology: Add one DHCP server, one switch, one router, and one PC. Connect them using cables.
2. Configure DHCP: On the server, open Services → DHCP and set:
   - Start IP: `192.168.1.10`
   - Subnet Mask: `255.255.255.0`
   - Default Gateway (incorrect): `192.168.1.254`
   - Maximum Users: `50`
   - Turn DHCP On and click Add.
3. Connect the client: On the PC, open Desktop → IP Configuration → DHCP.
4. Verify the gateway: Open Command Prompt and run:
   ipconfig
   The PC should display the gateway as `192.168.1.254`, even if the router's actual IP is `192.168.1.1`.

5. Identify and fix the issue: Compare the gateway shown on the PC with the router's actual LAN IP. Correct the DHCP pool's default gateway to `192.168.1.1`, renew the client's DHCP lease, and verify again.

Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

ANS:

DNS Troubleshooting Steps

1. Test connectivity by IP address.

ping 8.8.8.8

If successful, basic network connectivity is working.

2. Test DNS resolution using nslookup.

nslookup www.google.com

If it fails or shows a DNS timeout, the DNS server may be misconfigured or unavailable.

3. Check the configured DNS server.

ipconfig /all

Verify that the correct DNS server IP address is configured.

4. Flush the DNS cache.

ipconfig /flushdns

5. Fix the issue.

Correct the DNS server address in the network settings or DHCP configuration. For testing, use a valid DNS server such as `8.8.8.8`, if reachable and permitted.

6. Verify again.

nslookup www.google.com
ping www.google.com

A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

ANS:

DHCP Troubleshooting: APIPA Address (169.x.x.x)

Possible causes

1. DHCP server is switched off or unavailable.
2. Network cable or Wi-Fi connection is disconnected.
3. DHCP scope is exhausted (no free IP addresses).
4. Incorrect DHCP pool configuration.
5. DHCP requests are blocked or the relay agent is misconfigured.

Step-by-step resolution

Step 1: Check the IP address

ipconfig /all

If the device has a `169.254.x.x` address, it may not have received a DHCP lease.

Step 2: Check network connectivity

- Verify cables, switch ports, and Wi-Fi connection.
- Ensure the client and DHCP server are connected to the correct network.

Step 3: Check DHCP server

- Confirm the server is powered on and the DHCP service is enabled.
- Verify the IP pool, subnet mask, and available addresses.

Step 4: Renew the IP address

ipconfig /release
ipconfig /renew

Step 5: Check DHCP scope

- Ensure free IP addresses are available.
- Expand the pool if it is exhausted.

Step 6: Verify the result

ipconfig

Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.
ANS:

Common DNS and DHCP Issues

1. DNS Issues

DNS Issue	Symptoms	Possible Cause	Recommended Solution
DNS resolution failure	Websites do not open by name	Incorrect DNS server address	Configure a valid DNS server
Incorrect DNS record	Domain resolves to the wrong IP	Wrong A or CNAME record	Correct the DNS zone file
DNS server unavailable	DNS timeout in `nslookup`	Server stopped or network issue	Restart DNS service and check connectivity

2. DHCP Issues

DHCP Issue	Symptoms	Possible Cause	Recommended Solution
APIPA address	Device gets `169.254.x.x`	DHCP server unavailable	Check server and network connectivity
DHCP pool exhausted	New device cannot get an IP	No available IP addresses	Expand the DHCP pool

Incorrect default gateway	Local network works, internet does not	Wrong gateway in DHCP settings	Correct the gateway and renew the lease

DAY 1- DAY 5

Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

ANS:

Static IP Configuration for Five Devices

1. Assign static IP addresses

Configure each device under Network Settings → IPv4 Properties.

Device	IP Address	Subnet Mask
PC1	192.168.1.10	255.255.255.0
PC2	192.168.1.11	255.255.255.0
PC3	192.168.1.12	255.255.255.0
PC4	192.168.1.13	255.255.255.0

PC5	192.168.1.14	255.255.255.0

2. Connect the devices

Connect all five devices to the same switch or LAN. Ensure these IP addresses are not already in use.

3. Verify IP configuration

Open Command Prompt on each device and run:

ipconfig

4. Test connectivity

From PC1, run:

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

Expected result: Successful replies confirm connectivity between PC1 and the other devices. Repeat from other PCs if needed.

DHCP Server Setup — Cisco Packet Tracer

1. Create the topology: Add 1 server, 1 switch, and 3 PCs. Connect all devices to the switch.

2. Configure the server: Set the server IP to `192.168.1.1` and subnet mask to `255.255.255.0`.

3. Enable DHCP: Go to Server → Services → DHCP and configure:

- Service: On
- Pool Name: `LAN`

- Start IP: `192.168.1.10`
- Subnet Mask: `255.255.255.0`
- Maximum Users: `50`

Click Add.

4. Configure clients: On each PC, select Desktop → IP Configuration → DHCP.

5. Verify assigned IPs:

Device	Expected IP Address
PC0	192.168.1.10
PC1	192.168.1.11
PC2	192.168.1.12

6. Test connectivity: From PC0, open Command Prompt and run `ping 192.168.1.11` and `ping 192.168.1.12`.

Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

ANS:

Configure Custom DNS: `musicplayer.local`

1. Set up a local web server

Install Apache on a Linux machine and create a simple web page:

sudo apt install apache2 -y

echo "Welcome to Music Player" | sudo tee /var/www/html/index.html

Assume the web server IP is `192.168.1.50`.

2. Configure DNS using dnsmasq

Install dnsmasq:

sudo apt install dnsmasq -y

Add this entry to `/etc/dnsmasq.conf`:

address=/musicplayer.local/192.168.1.50

Restart the service:

sudo systemctl restart dnsmasq

3. Configure connected devices

Set the network's DNS server to the IP address of the machine running dnsmasq. Ensure the DHCP server distributes that DNS address to clients.

4. Test the hostname

Run:

nslookup musicplayer.local

Expected result: `192.168.1.50`

Then open a browser and visit:

`http://musicplayer.local`

Configure a DNS service (using your router’s DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

ANS:

Configure Local DNS for `musicplayer.local`

Install dnsmasq on Linux:
sudo apt update

1. sudo apt install dnsmasq apache2 -y
2. Create a local webpage:
   echo "Welcome to Music Player" | sudo tee /var/www/html/index.html
   Assume the server IP is `192.168.1.50`.
3. Add the DNS mapping:
   sudo nano /etc/dnsmasq.conf
   Add:
   address=/musicplayer.local/192.168.1.50
4. Restart DNS service:
   sudo systemctl restart dnsmasq
5. Configure client devices: Set their DNS server to the IP address of the dnsmasq machine.
6. Test the setup:
   nslookup musicplayer.local
   Then open `http://musicplayer.local` in a browser.

Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

ANS:

VLAN Configuration in Cisco Packet Tracer

1. Create the network topology

Add one Cisco 2960 switch and four PCs. Connect all PCs to the switch.

Device	VLAN	IP Address
PC0	VLAN 10 – Workstations	192.168.10.1
PC1	VLAN 10 – Workstations	192.168.10.2
PC2	VLAN 20 – Smart Devices	192.168.20.1
PC3	VLAN 20 – Smart Devices	192.168.20.2

Subnet mask for all devices: `255.255.255.0`

2. Configure the switch

Enter these commands in the switch CLI:

```
enable
configure terminal

vlan 10
name Workstations
exit

vlan 20
name Smart_Devices
```

exit

interface range fastEthernet 0/1-2

switchport mode access

switchport access vlan 10

exit

interface range fastEthernet 0/3-4

switchport mode access

switchport access vlan 20

exit

end

show vlan brief

3. Test connectivity

From each PC, open Desktop → Command Prompt.

- PC0 → PC1: `ping 192.168.10.2` — should succeed.
- PC2 → PC3: `ping 192.168.20.2` — should succeed.
- PC0 → PC2: `ping 192.168.20.1` — should fail.

Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

ANS:

NAT Configuration in Cisco Packet Tracer

1. Create the topology

Add one router, one switch, one PC, and one external server. Connect them appropriately.

2. Configure the router

Assume LAN network is `192.168.1.0/24` and the external network is `203.0.113.0/24`.

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

ip nat inside

no shutdown

exit

interface gigabitEthernet 0/1

ip address 203.0.113.1 255.255.255.0

ip nat outside

no shutdown

exit

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet 0/1 overload

end

3. Configure the PC

- IP address: `192.168.1.10`
- Subnet mask: `255.255.255.0`
- Default gateway: `192.168.1.1`

Configure the external server as `203.0.113.2/24` with gateway `203.0.113.1`.

4. Test connectivity

From the PC, open Desktop → Command Prompt and run:

ping 203.0.113.2

On the router, verify NAT translations:

show ip nat translations

show ip nat statistics